

LABORATORY OBSERVATION

Course	Full Stack Web Development
Course Code	23CM4121
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Experiment No.	PBL Part 2 – Task 7
Title	NPM and package.json – Installing and Using an External Package

1. TITLE

NPM and package.json – Installing and Using an External Package

2. AIM

To understand NPM and package.json and demonstrate the steps involved in installing an external package and using it in a Node.js application.

3. OBJECTIVE

- To understand the purpose of NPM.
 - To create and use package.json.
 - To install an external Node.js package.
 - To use the installed package in a Node.js program.
 - To understand how dependencies are recorded in package.json.
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4. THEORY

NPM (Node Package Manager) is used to install, manage, and share Node.js packages. The package.json file stores project metadata, scripts, and dependencies. In this task, the validator package is installed from NPM and used to validate an email address. The installed dependency is automatically recorded in package.json.

5. ALGORITHM / PROCEDURE

1. Create a project folder and open it in the terminal.
 2. Run `npm init -y` to create package.json.
 3. Install the external validator package using `npm install validator`.
 4. Create `app.js` and import validator.
 5. Read an email value in the program.
 6. Use `validator.isEmail()` to check whether the email is valid.
 7. Display the validation result.
 8. Run the application using `npm start`.
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6. PROGRAM

package.json

```
{
  "name": "npm-validator-demo",
  "version": "1.0.0",
  "description": "NPM and package.json demonstration",
  "main": "app.js",
  "scripts": {
    "start": "node app.js"
  },
  "dependencies": {
    "validator": "^13.15.20"
  }
}
```

```
}

app.js
const validator = require("validator");

const email = "student@example.com";

if (validator.isEmail(email)) {
  console.log(`${email} is a valid email address.`);
} else {
  console.log(`${email} is not a valid email address.`);
}
```

7. CODE EXPLANATION

Code	Explanation
npm init -y	Creates a default package.json file for the Node.js project.
npm install validator	Downloads the external validator package and records it as a dependency.
require('validator')	Imports the installed package into the application.
validator.isEmail(email)	Checks whether the supplied string has a valid email format.
npm start	Runs the start script defined in package.json.

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Test	Expected Result	Status
TC-01	student@example.com	Valid email message	Pass
TC-02	student@	Invalid email message	Pass

9. OUTPUT

Output 1: Program Output Screenshot

10. RESULT

Thus, NPM, package.json, dependency installation, and use of the external validator package were demonstrated successfully in a Node.js application.

How this would be submitted

Source Code

Task_7/

■■■ package.json

■■■ app.js

■■■ README.md